

Ameya Patil

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I design and build interactive data visualization and analysis systems for human-in-the-loop understanding of **large scale network graphs**, using an interdisciplinary approach of combining learnings from the fields of **Systems & Human Perception**.

I am also interested in **visualization perception research**, and spreading literacy through data visualizations, especially in the space of **socio-environmental** data contexts.

EXPERIENCE

University of Washington

Seattle WA, USA

PhD Student, advised by **Dr. Leilani Battle**

Sept 2021 - Present

Thesis: Evaluating Scalability and Interactivity in Network Visualization and Analysis Systems

- Performed a detailed survey and study of the state of the art in different tools available for managing, visualizing and analyzing network graph data, and synthesized future research directions in network data management and analysis. Work under review at IEEE InfoVIS 2026.
- Designed and implemented a benchmarking framework for evaluating graph systems on interactive network graph visualization and analysis workloads. Detected bugs in popular commercial graph systems using the benchmarking framework. Work under review at VLDB 2027.
- Performed a design study of domain experts in whale research to design WhaleVis - an interactive analysis dashboard for modeling and analyzing historical whale hunting data as a network graph. Published work in the International Whaling Commission Scientific Committee Meeting 2023 and IEEE InfoVIS 2023.

National Center for Atmospheric Research

Boulder CO, USA

Data Visualization Intern, advised by **Marlee Smith**, **Dr. Helen Kershaw** and **Dr. Moha El Ghararni**

Summer 2023

- Designed and implemented HydroVis - an interactive analysis dashboard for freshwater flood forecasting using the WRF-Hydro hydrological forecasting model. Published work as a demo in IEEE InfoVIS Viz4Climate + Sustainability Workshop, 2024.

AVIZ, Inria

Saclay, Paris, France

Research Intern, advised by **Dr. Jean-Daniel Fekete**

Summer 2021

- Proposed two new bar chart designs for progressive analytics and conducted a crowd-sourced user study to evaluate the new designs and the efficacy of confidence intervals for decision making in progressive analytics. Published work in TVCG 2023.

Fraunhofer CESE

College Park MD, USA

Research Assistant Intern, advised by **Dr. Marcel Schäfer**

Summer 2019

- Worked as Java developer on the PocketSecurity project which collects data to analyze user behaviour. Identified and implemented critical data probes to be collected for better analysis, and improved existing probes

NVIDIA

Pune MH, India

System Software Engineer - C/C++

July 2016 - July 2018

- Worked as developer for ShadowPlay - a gameplay sharing app to record, screenshot, broadcast and coplay video games, to develop new multi-threaded and multi-processes features, across software level code and GPU driver code.
- Enhanced and monitored the automated software testing suite for ShadowPlay and mentored an intern for the same.

NVIDIA

Pune MH, India

Intern

July 2015 - Dec 2015

- Device Filter Drivers - C/C++: Implemented end-to-end user input redirection from input devices to a specific application using filter drivers and device notifications
- Z-buffer - Python: Implemented aesthetic visual effects such as zoom burst using the depth data of images

SKILLS

- **Programming Languages:** C/C++, Java, Python, Javascript, SQL, Cypher, GSQL
- **Libraries/Frameworks:** D3.js, Matplotlib, Vega-lite, Altair, OpenCV, MPI, OpenMP
- **Soft Skills:** Collaboration, Public Speaking, Mentoring
- **Extra-Curricular:** Digital Photography, Photo and Video Editing

LANGUAGES

- **Marathi:** Native
- **Hindi:** Fluent
- **English:** Fluent
- **French:** Intermediate (B2)

EDUCATION

University of Washington, Seattle (UW)

Ph.D. in Computer Science, Advised by **Dr. Leilani Battle**, GPA: 4.00/4.00

WA, USA

Jan 2021 - Present

- **Thesis:** Evaluating Scalability and Interactivity in Network Visualization and Analysis Systems
- **Relevant Coursework:** Computing for Conservation, Computer Science Education seminar, Game Design

University of Maryland, College Park (UMD)

M.S. in Computer Science, GPA: 3.84/4.00

MD, USA

Aug 2018 - Dec 2020

- **Relevant Coursework:** Machine Learning, Geometric Computer Vision, Advanced Computer Graphics, Physically Based Modelling, Simulation & Animation, Interactive Data Analytics, Computational Geometry, Interactive Technologies in HCI, Database System Architecture and Implementation

Birla Institute of Technology and Science - Pilani (BITS)

B.E. (Honors) in Computer Science, GPA: 8.24/10.00

Goa, India

Aug 2012 - May 2016

- **Electives:** Data Mining, Data Storage Technologies and Networks, Creative Multimedia

PUBLICATIONS & WORKSHOP DEMOS

1. Z. Dong, T. Barrett, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "A Design Space of Behavior Change Interventions for Responsible Data Science", Proceedings of the Conference on Intelligent User Interface (IUI), 2025. DOI: 10.1145/3708359.3712140. [pdf](#)
2. Z. Dong, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "Behavior Matters: An Alternative Perspective on Promoting Responsible Data Science", Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), 2025. DOI: 10.1145/3710932. [arxiv:2410.17273](#)
3. **A. Patil**, M. Smith, H. Kershaw, M. El Gharamti, "Interactive Visualization of Ensemble Data Assimilation Forecasts for Freshwater Floods", IEEE VIS Viz4Climate + Sustainability Workshop Demo, 2024. [pdf](#)
4. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: Visualizing the History of Commercial Whaling", IEEE VIS Short Papers, 2023. DOI: 10.1109/VIS54172.2023.00028. [arxiv:2308.04552](#)
5. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: A New Visualization Tool for the IWC Catch Database", International Whaling Commission SC/69A/GDR/04, 2023. [archive.iwc.int/SC/69A/GDR/04](#)
6. **A. Patil**, G. Richer, C. Jermaine, D. Moritz, J.-D. Fekete, "Studying Early Decision Making for Progressive Bar Charts", IEEE Transactions on Visualization and Computer Graphics, 2023. DOI: 10.1109/TVCG.2022.3209426. HAL: <https://hal.science/hal-03738461/>
7. A. Aguinaldo, P.-Y. Chiang, A. Gain, **A. Patil**, K. Pearson and S. Feizi, "Compressing GANs using Knowledge Distillation", CoRR, vol. abs/1902.00159, 2019. [arXiv:1902.00159](#)